

Detecting Anomalies in Operational Data with OpenSearch

OpenSearch offers a robust and affordable anomaly detection solution. This fully open source and extensible platform incorporates machine learning capabilities straight into the operational data pipeline.



The scale, complexity, and velocity of contemporary IT infrastructures lead to a few challenges when using traditional DevOps methods. By incorporating artificial intelligence and machine learning into operations, AI-powered DevOps (AIOps) makes proactive monitoring, intelligent alerting, and autonomous remediation possible. An essential component of this ecosystem is the OpenSearch anomaly detection tool, which offers an open source, machine learning-based method for detecting anomalies in operational data in real time.

The role of AIOps

IT systems in the digital age are producing enormous volumes of data every second, ranging from events and logs to metrics and traces. Manually maintaining such systems is unsustainable and ineffective. Artificial intelligence for IT operations, or AIOps, automates and improves IT operations by using AI and machine learning, facilitating quicker

problem identification, diagnosis, and resolution.

Conventional IT monitoring tools are reactive, static, and rule based. As organisations adopt microservices, containers, hybrid cloud, and edge computing, IT environments have become complex, volatile and large. Intelligence, automation, and scalability are provided by AIOps to manage these environments.

Overview of OpenSearch

Originally based on Elasticsearch, OpenSearch is an open source platform for search, analytics, and observability. Maintained by the OpenSearch Project, which is headed by Amazon, it has been built to manage application performance monitoring (APM), full-text search, and log analytics. AIOps functions are supported by OpenSearch's machine learning (ML) plugins and observability features, especially for log-based anomaly identification, metrics analytics, and alerting.